

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 441012538

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.422325 d (ALIASED, see below)
-> corrected period: 4.844650 d
Epoch (t0) 1326.4091 BTJD
Duration 0.1200 d
Depth 4.261 % (42,608 ppm)
SNR 151.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

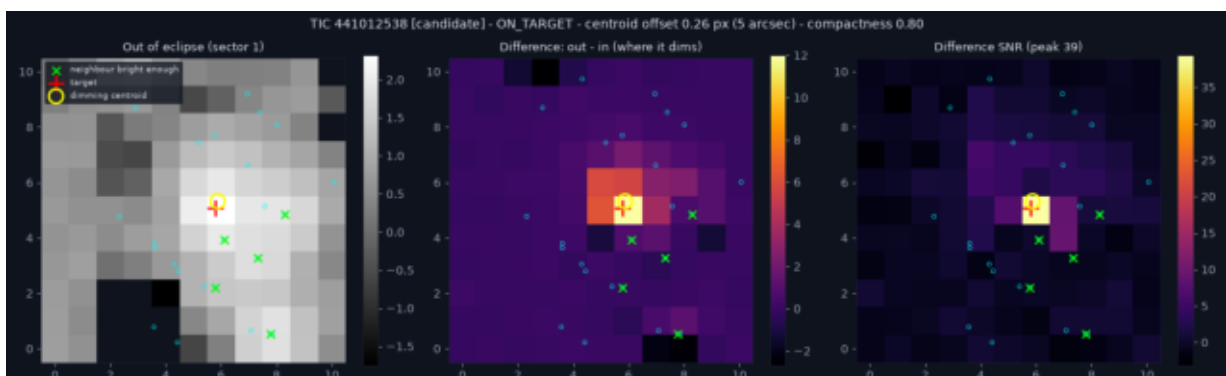
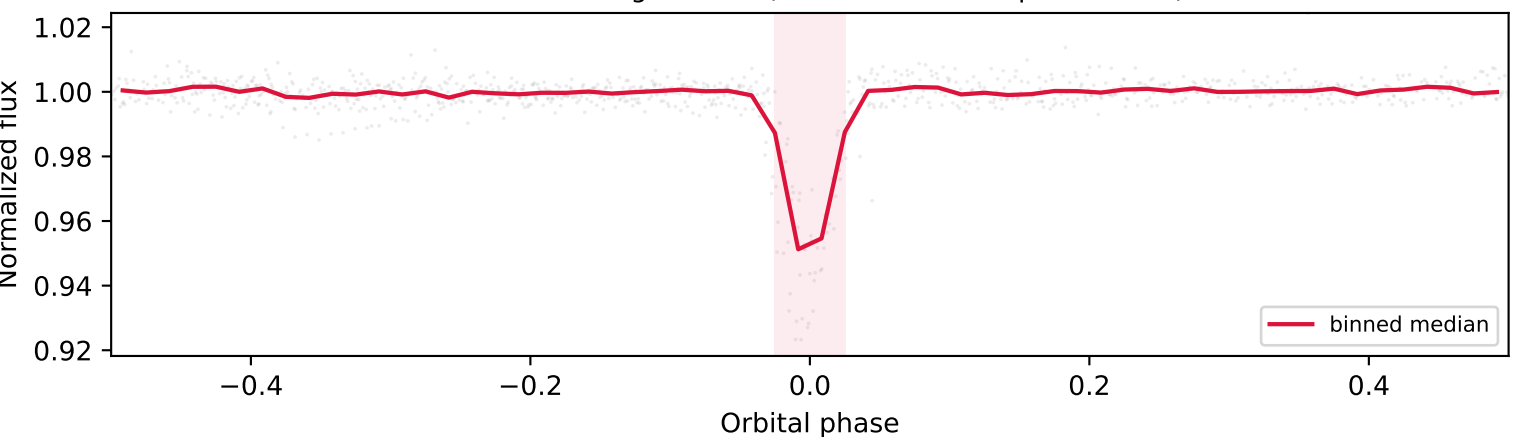
Eclipses observed 11 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 81.5 (gate ≥ 7.0)
Odd/even z-score 15.16 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 86.0 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 39.39 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.26 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 441012540
1.40 px away, dTmag +2.63 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 15.16 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.844650 d, in any write-up.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	20.362255 d (ALIASED, see below)	VSX	no match
-> corrected period:	40.724510 d	ASAS-SN	no match
Epoch (t0)	1750.0737 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1700 d	Gaia variable classification	no match
Depth	3.316 % (33,164 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	121.0	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	3 (gate >= 3)		
Robust depth ratio	1.10 (gate >= 0.50)		
Robust depth sigma	121.3 (gate >= 7.0)		
Odd/even z-score	8.62 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase -0.48, 5.5 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 16	light-curve dip in this sector: 9.7 sigma	t0 refined by	-0.4850 phase
Difference-image peak SNR	11.07 (gate >= 5.0)	Compactness	0.94 (gate >= 0.50)
Dimming centroid offset	0.70 px (15 arcsec)	(gate <= 1.0 px, i.e. on the target)	
Nearest bright-enough neighbour in cutout:	TIC 313386192		
	3.02 px away, dTmag +2.70 - farther from the dimming than the target		

